I followed the video below to install SQL Server 2019 using Docker:
https://www.youtube.com/watch?v=glxE7w4D8v8

The same approach can also be used to install later versions, such as SQL Server 2022. While the video demonstrates SQL Server 2019, the installation steps and Docker concepts are still relevant for newer versions, with only minor version-specific changes.

You are free to refer to any other YouTube video for installation as well, as there are many resources available that cover this process in detail.

I have installed SQL Server 2019 along with Azure Data Studio.

Query optimization techniques

mysql

sql server

I installed SQL server 2019 on my system
and then I installed Azure Data Studio
create database sql_week8;
Indexes
=========

Indexes help to speed up searching

orders table
==============
order_id (index)

select * from orders where order_id = ?

if there is no index on order_id

1 million orders in the orders table

select * from orders where order_id = ?

it would have to scan all the orders in this case

it has to go through 1 million rows to find it out.

it's like a linear search

A index (order_id)

1 million records

a btree structure

1000 elements you have to find a element

log 1000 = 10

1024
512
256
128
64
32
16
8
4
2
1

database
schema
table

```
CREATE TABLE orders(
order_id int NOT NULL,
 order_date datetime NOT NULL,
 order_customer_id int NOT NULL,
 order_status varchar(45) NOT NULL,
 PRIMARY KEY (order_id)
 );
```

```
insert into orders
select * from orders_temp;
```

I have created a primary key on order_id

order_id is unique + order_id will not hold null

also there will be a index on order_id

unique, clustered

clustered / nonclustered
unique / nonunique

```
select * from orders where order_id = 45;
```

check the estimated plan

clustered index seek

scan vs seek

scan is like scanning without index (a linear search)
seek is using the index (btree structure)

-- clustered index seek
select * from orders where order_id = 45;

estimated cpu - .0001

IO - .003

-- clustered index scan
select * from orders where order_customer_id = 45;
estimated cpu and IO cost

estimated cpu cost - .07

IO cost - .24

seek is generally much faster than scan.

when we create a primary key by default a index is created

(unique clustered index)
- clustered index
- nonclustered index